

Alireza Fadaei

Software Engineer · Backend Engineer

Oulu, Finland

Python · Node.js · Applied AI · Cybersecurity

Open to on-site, hybrid & remote opportunities

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arfadaei.ir · GitHub

LinkedIn · ORCID

Software and backend engineer based in Oulu, Finland, with experience building APIs, SaaS platforms, security software, research-data systems, and infrastructure using Python and Node.js. My background also includes applied AI, cybersecurity, federated learning, and peer-reviewed research.

EXPERIENCE

2026

Software Engineer · Aminyx (Remote)

Built agentic AI features for automated sourcing, multi-layered trust-scoring systems, and secure infrastructure for B2B smart contracts and international trade workflows, contributing to a modular AI-powered SaaS platform for global trade.

2024–2026

Programming Instructor · Iran TVTO

Delivered programming and software-development training through hands-on, industry-aligned projects in official vocational programmes.

2024–2026

Chair, Computer Department · Kabir Technical and Vocational School

Led the department team and developed curricula in programming, network security, and machine learning.

2022–2024

Software Developer · Fardaad (ISTT)

Built backend functionality for vulnerability and patch-management tools developed for municipal and government network environments. Improved backend algorithms supporting patch deployment workflows and collaborated through Git-based development and cross-functional review.

2021–2022

Backend Developer & Server Administrator · BrassicaLNC

Built the backend and administered a research-data platform organising data from 2,361 publicly available RNA-Seq libraries.

2021–2022

Backend Developer & Server Administrator · MetaMenu

Developed backend features and administered infrastructure for a cloud-based restaurant platform spanning interactive AR menus, ordering, delivery, and accounting workflows within an agile startup team.

2021–2022

Backend Developer · Piranest Team

Developed marketplace backend functionality connecting travellers with local tour guides and supporting tour reservations.

2020

Flask Developer · Fapsco Project

Developed and maintained Flask-based website features and internal tools, supporting stable client-facing services and day-to-day business workflows.

EDUCATION

2026–Present	M.Sc. Software Engineering and Information Systems University of Oulu · Oulu, Finland Building on my previous M.Sc. in Software Engineering, the programme extends my expertise in advanced software engineering, software quality, AI-enabled systems, and human-centred information systems.
2022–2025	M.Sc. Software Engineering Islamic Azad University, Najafabad Branch · Top student Thesis: Optimized Federated Deep Learning for Intrusion Detection in Dew Computing Environments
2017–2022	B.Sc. Computer Engineering University of Isfahan · Best B.Sc. Thesis Award Thesis: Design and Implementation of an ADS-B-Based Aircraft Observation System

SELECTED PUBLICATION

2026	A Novel Intrusion Detection System for Dew Computing Environments Based on an Enhanced Federated Deep Learning Model Alireza Fadaei, Behrang Barekatain · <i>The Journal of Supercomputing</i> , 2026 · Published February 16, 2026 · DOI: 10.1007/s11227-026-08314-x
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Research interests: Federated and distributed machine learning · Intrusion detection · Cybersecurity · Privacy-sensitive systems · Reliable AI-enabled software

TECHNICAL SKILLS

Backend Engineering	Python · Node.js · Django · Flask · FastAPI · REST APIs · Celery
Databases & Infrastructure	PostgreSQL · Redis · Neo4j · Docker · Nginx · Linux
Applied AI & Machine Learning	TensorFlow · Scikit-learn · Pandas · NumPy · Federated Learning · CNNs
Security & Distributed Systems	Vulnerability Management · Intrusion Detection · AI Security · IoT Security
Engineering Practices	Git · Code Review · Technical Documentation · Agile Development · API Design
Additional Technologies: VMware · React · Modern JavaScript	

LANGUAGES

English: Professional working proficiency · IELTS Academic: Overall 7.5, March 2025